

Handbook of Industrial Drying

Fourth Edition

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recondensation of moisture to the exposed surface. An appreciation of this internal movement of moisture is important when it is the controlling factor, as it occurs after the critical moisture content, in a drying operation carried to low final moisture contents. Variables such as air velocity and temperature, which normally enhance the rate of surface evaporation, are of decreasing importance except to promote the heat transfer rates. Longer residence times, and, where permissible, higher temperatures become necessary. In the case of such materials as ceramics and timber, in which considerable shrinkage occurs, excessive surface evaporation sets up high moisture gradients from the interior toward the surface, which is liable to cause overdrying, excessive shrinkage, and, consequently, high tension, resulting in cracking or warping. In such cases, it is essential not to incur too high moisture gradients by retarding surface evaporation through the employment of high air relative humidities while maintaining the highest safe rate of internal moisture movement by virtue of heat transfer. The temperature gradient set up in the solid will also create a vapor–pressure gradient, which will in turn result in moisture vapor diffusion to the surface; this will occur simultaneously with liquid moisture movement.

1.3.1 MOISTURE CONTENT OF SOLIDS

The moisture contained in a wet solid or liquid solution exerts a vapor pressure to an extent depending upon the nature of moisture, the nature of solid, and the temperature. A wet solid exposed to a continuous supply of fresh gas continues to lose moisture until the vapor pressure of the moisture in the solid is equal to the partial pressure of the vapor in the gas. The solid and gas are then said to be in *equilibrium*, and the moisture content of the solid is called the *equilibrium moisture content* under the prevailing conditions. Further exposure to this air for indefinitely long periods will not bring about any additional loss of moisture. The moisture content in the solid could be reduced further by exposing it to air of lower relative humidity. Solids can best be classified as follows [12]:

Nonhygroscopic capillary-porous media, such as sand, crushed minerals, nonhygroscopic crystals, polymer particles, and some ceramics. The defining criteria are as follows. (1) There is a clearly recognizable pore space; the pore space is filled with liquid if the capillary-porous medium is completely saturated and is filled with air when the medium is completely dry. (2) The amount of physically bound moisture is negligible; that is, the material is nonhygroscopic. (3) The medium does not shrink during drying.

Hygroscopic-porous media, such as clay, molecular sieves, wood, and textiles. The defining criteria are as follows. (1) There is a clearly recognizable pore space. (2) There is a large amount of physically bound liquid. (3) Shrinkage often occurs in the initial stages of drying. This category was further classified into (a) hygroscopic capillary-porous media (micropores and macropores, including bidisperse

media, such as wood, clays, and textiles) and (b) strictly hygroscopic media (only micropores, such as silica gel, alumina, and zeolites).

Colloidal (nonporous) media, such as soap, glue, some polymers (e.g., nylons), and various food products. The defining criteria are as follows: (1) there is no pore space (evaporation can take place only at the surface); (2) all liquid is physically bound.

It should be noted that such classifications are applicable only to homogeneous media that could be considered as continua for transport.

As a wet solid is usually swollen compared with its condition when free of moisture and its volume changes during the drying process, it is not convenient to express moisture content in terms of volume. The moisture content of a solid is usually expressed as the moisture content by weight of bone-dry material in the solid, X . Sometimes a wet basis moisture content W , which is the moisture–solid ratio based on the total mass of wet material, is used. The two moisture contents are related by the expression

$$X = \frac{W}{1 - W} \quad (1.50)$$

Water may become bound in a solid by retention in capillaries, solution in cellular structures, solution with the solid, or chemical or physical adsorption on the surface of the solid. Unbound moisture in a hygroscopic material is the moisture in excess of the equilibrium moisture content corresponding to saturation humidity. All the moisture content of a nonhygroscopic material is unbound moisture. Free moisture content is the moisture content removable at a given temperature and may include both bound and unbound moisture.

In the immediate vicinity of the interface between free water and vapor, the vapor pressure at equilibrium is the saturated vapor pressure. Very moist products have a vapor pressure at the interface almost equal to the saturation vapor pressure. If the concentration of solids is increased by the removal of water, then the dissolved hygroscopic solids produce a fall in the vapor pressure due to osmotic forces. Further removal of water finally results in the surface of the product dried. Water now exists only in the interior in very small capillaries, between small particles, between large molecules, and bound to the molecules themselves. This binding produces a considerable lowering of vapor pressure. Such a product can therefore be in equilibrium only with an external atmosphere in which the vapor pressure is considerably decreased.

1.3.2 MOISTURE ISOTHERMS [10]

A dry product is called hygroscopic if it is able to bind water with a simultaneous lowering of vapor pressure. Different products vary widely in their hygroscopic properties. The reason for this is their molecular structure, their solubility, and the extent of reactive surface.

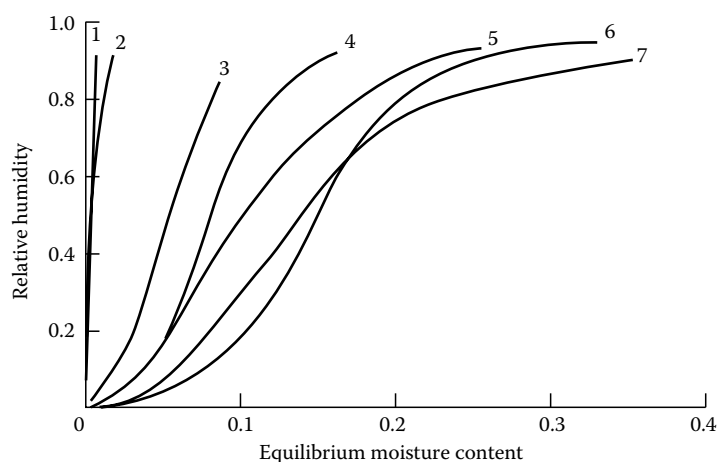


FIGURE 1.6 Typical equilibrium moisture isotherms at room temperature for selected substances: (1) asbestos fiber, (2) PVC (50°C), (3) wood charcoal, (4) Kraft paper, (5) jute, (6) wheat, (7) potatoes.

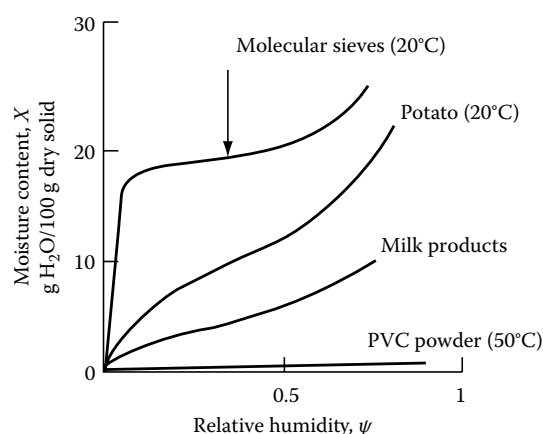


FIGURE 1.7 Shapes of sorption isotherms for materials of varying hygroscopicity.

Sorption isotherms measured experimentally under isothermal conditions are used to describe the hygroscopic properties of a product. A graph is constructed in which the moisture bound by sorption per unit weight is plotted against relative humidity, and vice versa. Such isotherms are shown in Figures 1.6 and 1.7. From Figure 1.7 it is seen that molecular sieves are highly hygroscopic but polyvinyl chloride (PVC) powder is mildly hygroscopic. Potatoes and milk exhibit intermediate hygroscopicity.

Figure 1.8 shows the shape of the sorption isotherm characteristic of many dry food products. If the partial pressure of the external atmosphere P_w is nearly zero, then the equilibrium moisture inside the dry product will also be almost zero. Section A of the curve represents a region in which the monomolecular layers are formed, although there may be multimolecular layers in some places toward the end of A. Section B is a transitional region in which double and multiple layers are mainly formed. Capillary condensation could also have taken place. In section C the slope of the curve increases again, which is attributed mainly to increasing capillary condensation and swelling. The maximum hygroscopicity

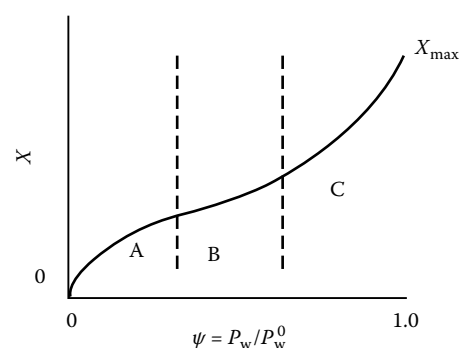


FIGURE 1.8 A typical isotherm (see text for explanation of areas within dashed lines).

$X_{\max}$ is achieved when the solid is in equilibrium with air saturated with moisture ($\psi = 1$).

1.3.2.1 Sorption–Desorption Hysteresis

The equilibrium moisture content of a product may be different depending on whether the product is wetted (sorption or absorption) or dried (desorption) (Figure 1.9). These differences are observed to varying degrees in almost all hygroscopic products.

One of the hypotheses used to explain hysteresis is to consider a pore connected to its surroundings by a small capillary [10]. During absorption, as the relative humidity rises, the capillary begins to fill while the pore is empty. Only when the partial pressure of the vapor in air is greater than the vapor pressure of the liquid in the capillary will the moisture move into the pore. Starting from saturation the pore is full of liquid. This fluid can only escape when the partial pressure of the surrounding air falls below the vapor pressure of the liquid in the capillary. Since the system of pores has generally a large range of capillary diameters, it follows that differences between adsorption and desorption will be observed. This theory assumes that the pore is a rigid structure. This is not true for foods or synthetic materials, although these show hysteresis. The explanation is that contraction and swelling

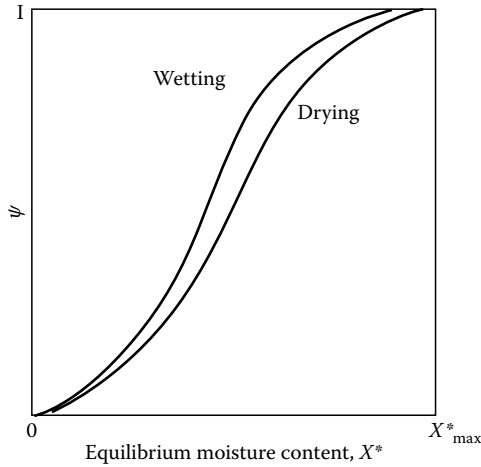


FIGURE 1.9 Wetting and drying isotherms for a typical hygroscopic solid.

are superimposed on the drying and wetting processes, producing states of tension in the interior of the products and leading to varying equilibrium moisture contents depending on whether desorption or absorption is in progress.

1.3.2.2 Temperature Variations and Enthalpy of Binding

Moisture isotherms pertain to a particular temperature. However, the variation in equilibrium moisture content for small changes of temperature ($<10^{\circ}\text{C}$) is neglected [3]. To a first approximation, the temperature coefficient of the equilibrium moisture content is proportional to the moisture content at a given relative humidity:

$$\left(\frac{\partial X^*}{\partial T} \right)_{\psi} = -AX^* \quad (1.51)$$

The coefficient A lies between 0.005 and 0.01 per kelvin for relative humidities between 0.1 and 0.9 for such materials as natural and synthetic fibers, wood, and potatoes. A could be taken to increase linearly with ψ . So for $\psi = 0.5$ there is a 0.75% fall in moisture content for each degree kelvin rise in temperature. The extent of absorption–desorption hysteresis becomes smaller with increasing temperature.

Figure 1.10 shows moisture isotherms at various temperatures. The binding forces decrease with increasing temperature; that is, less moisture is absorbed at higher temperatures at the same relative humidity. Kessler [10] has shown that the slope of a plot of $\ln(P_w/P_w^0)$ versus $1/T$ at constant X (Figure 1.11) gives the enthalpy of binding. The variation of enthalpy of binding versus moisture content is shown in Figure 1.12. From the figure it is seen that in the region where monomolecular layers are formed, enthalpies of binding are very high.

1.3.3 DETERMINATION OF SORPTION ISOTHERMS [10]

The sorption isotherms are established experimentally starting mostly with dry products. The initial humidity of the air with which the product is in equilibrium should be brought

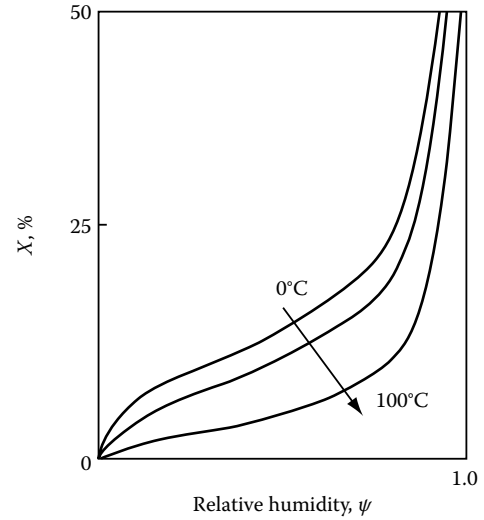


FIGURE 1.10 Sorption isotherms for potatoes.

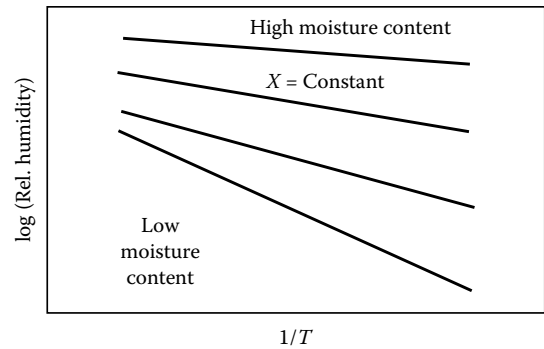


FIGURE 1.11 Determination of the heat of sorption from sorption isotherms.

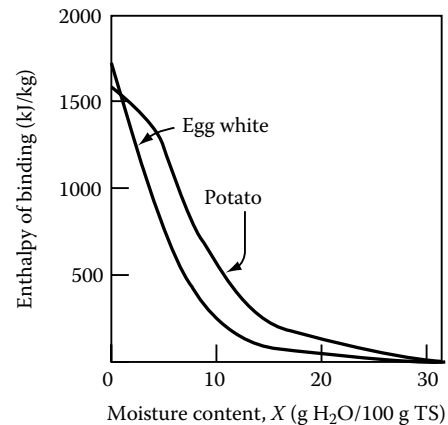


FIGURE 1.12 Enthalpy of sorption as a function of the hygroscopic moisture content. (Egg white data by Nemitz; potato data by Krischer.)